Further applications of the normal distribution

1 The operating times of phone batteries are approximately normally distributed with a mean of 34 hours and a standard deviation of 4 hours. Answer the following questions using the empirical rule:

- **a** What percentage of batteries last between 33 and 38 hours? Round your answer to the nearest percent.
- **b** What percentage of batteries last between 28 hours and 41 hours?
- **c** Any battery that lasts less than 23 hours is deemed faulty. If a company manufactured $51\,000$ batteries, approximately how many batteries would they be able to sell? Round your answer to the nearest integer.

2 The height of sunflowers is approximately normally distributed, with a mean height of $1.6\ \text{m}$ and a standard deviation of $8\ \text{cm}$.

- **a** Approximately what percentage of sunflowers are between $1.55\ \text{m}$ and $1.60\ \text{m}$ tall? Round your answer to the nearest percent.
- **b** Approximately what percentage of sunflowers are between $1.69\ \text{m}$ and $1.81\ \text{m}$ tall? Round your answer to the nearest percent.
- **c** If there are 3000 sunflowers in the field, approximately how many are taller than $1.52\ \text{m}$? Round your answer to the nearest integer.

3 In the Maths Methods course at Winter Heights High, the mean mark for the year was 62% and the standard deviation was 13%.

- **a** Sophia's mark for the year was 72. Calculate her z-score.
- **b** Yuri's z-score for the year was 0.62. What was his actual percentage score?
- **c** Those students in or below the 0.01 quantile are advised that this subject is unlikely to count towards their ATAR calculation. Using a CAS calculator or otherwise, find what mark students need to score above in order for Methods to be used for their ATAR calculation.

4 The length of the tail of a domestic cat is normally distributed with a mean of $25\ \text{cm}$ and a standard deviation of $2.2\ \text{cm}$. Use your CAS calculator to answer the following, rounding your answers to three decimal places:

- **a** What is the shortest length of a tail in the 70th percentile?
- **b** What is the shortest length of a tail in the top 15%?
- **c** What is the shortest length of a tail in the 0.45 quantile?
- **d** What is the probability that a cat has a tail length less than $23.5\ \text{cm}$?
- **e** Suppose that a cat has a tail length below the 80th percentile. What is the probability that their tail length is more than $23.5\ \text{cm}$?

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5 The weights of babies born in a hospital in Sydney are considered normally distributed with a mean of $3.3\ \text{kg}$ and a standard deviation of $0.50\ \text{kg}$. Use your CAS calculator to answer the following, rounding your answers to three decimal places:

a What is the weight of the smallest baby in the 65th percentile?

b What is the weight of the smallest baby in the top 20%?

c What is the weight of the smallest baby in the 0.55 quantile?

d What is the probability that a baby weighs less than $3.05\ \text{kg}$? Give your answer as a decimal.

e Suppose that a baby has a weight below the 70th percentile. What is the probability that their weight is more than $3.05\ \text{kg}$?

6 People in the Dinaric Alps are considered the tallest in the world. The heights of males are considered to be normally distributed with a mean height of $186.0\ \text{cm}$ and a standard deviation of $6\ \text{cm}$. Use your CAS calculator to answer the following, rounding your answers to three decimal places unless otherwise stated:

a What is the height of the shortest male in the top 25%?

b What is the height of the tallest male in the 0.95 quantile?

c What percentage of males are shorter than $176\ \text{cm}$? Round your answer to the nearest tenth of a percent.

d What is the probability that a male has a height less than $187\ \text{cm}$?

e Suppose that a male has a height below the 90th percentile. What is the probability that their height is more than $187\ \text{cm}$?

7 A machine is set for the production of cylinders of mean diameter $5.06\ \text{cm}$, with standard deviation $0.016\ \text{cm}$.

a Assuming a normal distribution, between what values, in centimeters, will 99.7% of the diameters lie?

b If cylinders with diameters less than $5.012\ \text{cm}$ or more than $5.108\ \text{cm}$ are discarded, what percentage of cylinders produced are discarded?

c If a cylinder, randomly selected from this production, has a diameter of $5.124\ \text{cm}$, what conclusion could be drawn?

8 A weather station records temperatures normally distributed with a mean of 28°C and a standard deviation of 3.3°C. The temperatures were converted to Kelvin using the rule $Y = X + 273.15$ where X is the temperature in Celsius and Y is the temperature in Kelvin.

a Find the new mean.

b Find the new standard deviation.

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9 The weights of babies born in a hospital in Sydney are considered normally distributed with a mean of $3.5\ \text{kg}$ and a standard deviation of $0.55\ \text{kg}$. If the weights were recorded in grams instead of in kilograms:

a Find the new mean.

b Find the new standard deviation.

10 The heights of a certain species of fully grown plants are thought to be normally distributed with a mean of $55\ \text{cm}$ and a standard deviation of $4\ \text{cm}$. If the heights were recorded in millimetres instead of centimetres:

a Find the new mean.

b Find the new variance.

11 The marks in the Chemistry ATAR exam were normally distributed. Let $X\%$ be the random variable representing the distribution of these marks. Dylan scored a raw mark of 57% and after average marks scaling scored 63.28%. Danielle scored a raw mark of 86% and after average marks scaling scored 93.44%.

a If these marks were scaled according to the rule $aX + b$, find the values of a and b.

b If the raw mean mark was 62%, find the scaled mean mark for Chemistry.

12 In a given population, a certain variable X is considered to be normally distributed with a mean of 80 and a standard deviation of 4. If the data for Y is transformed according to the rule $-8 - 4X$, calculate the new:

a Mean

b Standard deviation

13 The weights of extra large free range eggs are normally distributed such that the mean weight is $68\ \text{g}$ and the standard deviation is $0.67\ \text{g}$.

a Before using an egg in a recipe, Roald weighs the egg and finds it weighs $70\ \text{g}$. Find the z-score for this egg, rounded to three decimal places.

b The eggs are advertised as weighing $70\ \text{g}$ each. Tina weighs each of the eggs in her dozen and finds that all of them are in the 41st quantile. She decides to take them back for a refund. Using a CAS calculator or otherwise, calculate the mass of a single egg in her carton.

14 The full height of dwarf lemon trees are normally distributed with a mean of $69\ \text{cm}$ and a standard deviation of $\sigma\ \text{cm}$. It is known that 20% of plants are smaller than $40\ \text{cm}$.

a Using a CAS calculator or otherwise, calculate the standard deviation of these plant heights correct to three decimal places.

b Let X represent the distribution of the dwarf lemon tree plant heights. If a new variety of plant is being developed to be $11\ \text{cm}$ taller, write a rule in terms of X that can be used to scale the heights of the currents plants.

c Hence, calculate the standard deviation of the new variety of lemon tree correct to three decimal places.

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15 A random variable X is normally distributed with a mean of μ and a standard deviation of σ.

a It is known that 76% of scores lie below 70. Use the standardisation formula to write a rule in terms of μ and σ. Round the z-value to three decimal places.

b After a scaling of $X + 12$, it is known that 63% of scores lie above 56. Write another rule in terms of μ and σ using this information and round the z-value to three decimal places.

c Use the simultaneous solving facility on your calculator to find the values of μ and σ.

16 The marks in the Maths Methods ATAR exam were normally distributed and had a raw mean mark of μ and a standard deviation of σ. The exam was out of 150. After average marks scaling, all the marks were scaled up by 5% and then an extra 3 marks were added.

a If a student scored 81 marks, what would their new mark be, as a percentage, after scaling?

b Write an expression for the new mean mark as a percentage in terms of μ.

c Write an expression for the new standard deviation as a percentage in terms of σ.

17 The marks for an IQ test are normally distributed with a mean of μ and a standard deviation of 10. It is known that 13% of participants score above 134 marks.

a Using a CAS calculator or otherwise, calculate the mean mark for the IQ test correct to three decimal places.

b Let X represent the distribution of the IQ test results. If the test is out of 185, write a rule in terms of X to scale all the results to a percentage.

18 At Summer Heights High, 148 students are studying Physics. The results on the first semester examination saw a mean mark of 60% with a standard deviation of 11.4%, and the results were considered to be approximately normally distributed.

a Luke scored 63% on the first semester exam. What percentage of students scored more than Luke on the first semester exam? Round your answer to the nearest percentage.

b Students whose score was more than two standard deviations below the mean were advised not to continue with the study of Physics. At most, how many students were given this advice?

c The results of the end of year examination saw a mean mark of 67% with a standard deviation of 9.3%. If Luke did just as well in the end of year examination, relative to his peers, as he did in the first semester examination, then what score would Luke receive?

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19 Victoria downloads each episode of her favourite TV show as it's released online. The length of each show is represented by the random variable T, which is approximately normally distributed with a mean length of 50 minutes and a standard deviation of 4 minutes.

a What percentage of her shows are less than 49 minutes in length?

b Victoria wants to put a show on her USB drive but only has room for an episode that is 48 minutes in length. What is the probability that she won't be able to fit the show on the drive?

c Of the next five shows that Victoria independently downloads, what is the probability that the first two are less than 49 minutes and the last three are more than 49 minutes?

d Of the next 5 shows that Victoria downloads, what is the probability that exactly two are less than 49 minutes?

e Fans of the show have complained that the show length is really inconsistent. Calculate the maximum value of the standard deviation such that the probability of a show being less than 45 minutes is no more than 0.2%.

20 The raw exam results, X, for an ATAR subject is approximately normally distributed with a mean of 56% and a standard deviation of 13%.

a If 2700 students sat the exam, what is the lowest score corresponding to the top 54 scoring students?

b If 14 results are chosen at random for this subject, what is the probability that at most 5 students scored less than 51%?

c The examination committee wish to maintain the spread of results but alter the mean so that less than 5% of students scored less than 23%. Determine the value of the scaled mean mark.

21 A packet of crisps is advertised as weighing 175 g, but the actual packets are normally distributed with a mean mass of 174.8 g and a standard deviation of 1.2 g.

a What is the probability that a randomly selected packet weighs more than the advertised weight?

b What is the largest weight out of the lightest 5% of packets?

c Out of 11 randomly selected packets, what is the probability that at least 5 packets weigh more than 174.6 g?

d The company decides that no more than 3% of packets should be underweight. They increase the mean weight of the bags without changing the standard deviation. What does the new mean weight need to be to meet this condition?

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